Linear Algebra: Abstract Structures, Eigentheory, and Spectral Analysis

Advanced Course Notes

Abstract Vector Spaces

A vector space over a field F is a set V together with two operations, addition $+ : V \times V \to V$ and scalar multiplication $\cdot : F \times V \to V$, satisfying eight axioms. For $u, v, w \in V$ and $a, b \in F$:

1. $u + v = v + u$ (commutativity).
2. $(u + v) + w = u + (v + w)$ (associativity).
3. There exists $\mathbf{0} \in V$ such that $v + \mathbf{0} = v$ for all v.
4. For each v there exists $-v$ such that $v + (-v) = \mathbf{0}$.
5. $1 \cdot v = v$ (identity scalar).
6. $a(bv) = (ab)v$ (associativity of scalar multiplication).
7. $a(u + v) = au + av$ (distributivity over vector addition).
8. $(a + b)v = av + bv$ (distributivity over scalar addition).

The key insight is that V need not consist of tuples of numbers. Polynomials of degree at most n, continuous functions on an interval, infinite sequences, and matrices of fixed size are all vector spaces under appropriate operations. The axioms capture the structural essence of "linear combination" divorced from any particular representation.

A subspace $W \leq V$ is a subset closed under addition and scalar multiplication (equivalently, closed under all linear combinations). The empty set fails to be a subspace since it lacks $\mathbf{0}$; the canonical subspaces are $\{\mathbf{0}\}$ and V itself.

The intersection of any family of subspaces is a subspace. The sum $W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}$ is a subspace. If $W_1 \cap W_2 = \{\mathbf{0}\}$, the sum is direct: $V = W_1 \oplus W_2$, and every $v \in V$ decomposes uniquely as $w_1 + w_2$.

Linear Independence, Span, and Dimension

A set $S \subseteq V$ is linearly independent if no finite linear combination $\sum_i c_i v_i = \mathbf{0}$ (with $v_i \in S$) is possible except with all coefficients zero. This is equivalent to: no element of S lies in the span of the others.

The span of S is the smallest subspace containing S: all finite linear combinations of elements of S. A set S that is both independent and spanning for V is a basis. By the replacement theorem (or Steinitz exchange lemma), any two bases of V have the same cardinality, called the dimension $\dim V$.

In a finite-dimensional space of dimension n:

- Any independent set has at most n elements; if it has exactly n, it is a basis.
- Any spanning set has at least n elements; if it has exactly n, it is a basis.
- Any n independent vectors automatically span V.

The coordinate isomorphism: once a basis $\mathcal{B} = (b_1, \ldots, b_n)$ is fixed, every $v = \sum_i c_i b_i$ is represented by its coordinate vector $[v]_{\mathcal{B}} = (c_1, \ldots, c_n)^T \in \mathbb{R}^n$. This isomorphism is canonical given $\mathcal{B}$ but depends on the ordering.

Linear Maps: Kernel, Image, Rank-Nullity

A linear map $T : V \to W$ between vector spaces satisfies $T(au + bv) = aT(u) + bT(v)$ for all $u, v \in V$ and $a, b \in F$. Linear maps are morphisms in the category of vector spaces: they preserve the algebraic structure.

The kernel $\ker T = \{v \in V : T(v) = \mathbf{0}\}$ is a subspace of V. The image $\operatorname{im} T = \{T(v) : v \in V\}$ is a subspace of W.

T is injective if and only if $\ker T = \{\mathbf{0}\}$. T is surjective if and only if $\operatorname{im} T = W$. A bijective linear map is an isomorphism.

The rank-nullity theorem: for $T : V \to W$ with V finite-dimensional,

$$\dim \ker T + \dim \operatorname{im} T = \dim V.$$

This fundamental identity constrains how a linear map can distribute the dimension of its domain between what it kills and what it produces. It implies that T cannot be injective and fail to be surjective simultaneously when $\dim V = \dim W$.

The matrix of T with respect to bases $\mathcal{B}$ (for V) and $\mathcal{C}$ (for W) is the $m \times n$ matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ whose j-th column is $[T(b_j)]_{\mathcal{C}}$. Changing bases replaces this matrix by $Q^{-1}[T]_{\mathcal{B}}^{\mathcal{C}}P$, where P and Q are the change-of-basis matrices.

Eigenvalues and Eigenvectors: Abstract Formulation

Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V over F. A scalar $\lambda \in F$ is an eigenvalue of T if there exists a nonzero $v \in V$ with

$$T(v) = \lambda v.$$

The nonzero vectors satisfying this are the eigenvectors corresponding to λ. The eigenspace is

$$E_\lambda = \ker(T - \lambda I) = \{v \in V : T(v) = \lambda v\},$$

which includes $\mathbf{0}$ and is a nonzero subspace of V.

The eigenvalue equation $T(v) = \lambda v$ can be rewritten as $(T - \lambda I)(v) = \mathbf{0}$. An eigenvalue exists precisely when $T - \lambda I$ is not injective, i.e., when $\ker(T - \lambda I) \neq \{\mathbf{0}\}$.

In coordinates, if $A = [T]_\mathcal{B}$ is the matrix of T in some basis, then λ is an eigenvalue of T if and only if

$$\det(A - \lambda I) = 0.$$

The eigenvalues are coordinate-free invariants of T: they do not depend on the choice of basis, since a change of basis replaces A by $P^{-1}AP$, and $\det(P^{-1}AP - \lambda I) = \det(P^{-1}(A - \lambda I)P) = \det(A - \lambda I)$.

The Characteristic Polynomial

The characteristic polynomial of T (or of its matrix A) is

$$p_A(\lambda) = \det(A - \lambda I) = (-1)^n \lambda^n + \cdots + \det(A).$$

This is a degree-n polynomial. Its leading coefficient is $(-1)^n$ (for the $\det(-\lambda I)$ term). Two notable coefficients:

- Constant term ($\lambda = 0$): $p_A(0) = \det(A)$.
- Coefficient of λ^{n-1}: up to sign, equals $\operatorname{tr}(A)$, the trace (sum of diagonal entries).

By the factor theorem, the roots of $p_A(\lambda)$ are the eigenvalues. Over $\mathbb{C}$, the fundamental theorem of algebra guarantees n roots (counted with multiplicity). Over $\mathbb{R}$, some eigenvalues may be complex.

The Cayley-Hamilton theorem states that every matrix satisfies its own characteristic polynomial: $p_A(A) = 0$. This is a remarkable result: substituting the matrix itself into the polynomial gives the zero matrix. It has practical implications, including that A^{-1} can be expressed as a polynomial in A (of degree at most $n - 1$) when A is invertible.

The algebraic multiplicity $m_a(\lambda)$ of an eigenvalue is its multiplicity as a root of p_A. The geometric multiplicity $m_g(\lambda) = \dim E_\lambda = \dim \ker(A - \lambda I)$ satisfies $1 \leq m_g(\lambda) \leq m_a(\lambda)$.

Diagonalization and the Spectral Decomposition

A linear operator $T : V \to V$ is diagonalizable if there exists a basis of V consisting entirely of eigenvectors. In this eigenbasis, the matrix of T is diagonal:

$$[T]_\mathcal{E} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

In terms of matrices: A is diagonalizable if there exists an invertible P with $P^{-1}AP = D$ diagonal.

Diagonalizability criteria:

- A is diagonalizable iff $m_g(\lambda) = m_a(\lambda)$ for every eigenvalue λ.
- Equivalently, the eigenspaces together have total dimension n.
- If A has n distinct eigenvalues, it is automatically diagonalizable (since eigenvectors for distinct eigenvalues are independent).

Eigenvectors corresponding to distinct eigenvalues are linearly independent. Proof: suppose $T(v_i) = \lambda_i v_i$ for distinct λ_i and $\sum c_i v_i = 0$. Apply $T - \lambda_1 I$ to eliminate v_1, giving $\sum_{i>1} c_i(\lambda_i - \lambda_1)v_i = 0$. Repeating eliminates vectors one by one; since the λ_i are distinct and the v_i are nonzero, all $c_i = 0$.

The spectral decomposition: if $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, writing $P = [v_1 \mid \cdots \mid v_n]$ and $P^{-1} = \begin{bmatrix} w_1^T \\ \vdots \\ w_n^T \end{bmatrix}$:

$$A = \sum_{i=1}^{n} \lambda_i v_i w_i^T.$$

Each $v_i w_i^T$ is a rank-one projector onto the eigenspace of λ_i along the complementary space.

Vector Spaces with Inner Products

An inner product on a real vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

1. Symmetry: $\langle u, v \rangle = \langle v, u \rangle$.
2. Linearity in first argument: $\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$.
3. Positive definiteness: $\langle v, v \rangle \geq 0$ with equality iff $v = \mathbf{0}$.

The induced norm is $\|v\| = \sqrt{\langle v, v \rangle}$. The Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \|u\|\|v\|$, with equality iff u and v are proportional. The triangle inequality follows: $\|u + v\| \leq \|u\| + \|v\|$.

Orthogonality in the inner product sense: $u \perp v$ iff $\langle u, v \rangle = 0$. An orthonormal basis $\{q_1, \ldots, q_n\}$ satisfies $\langle q_i, q_j \rangle = \delta_{ij}$. In orthonormal coordinates, inner products and matrix representations simplify considerably.

The Gram-Schmidt process: given a basis $\{v_1, \ldots, v_n\}$, produce orthonormal $\{q_1, \ldots, q_n\}$ by

$$u_k = v_k - \sum_{j=1}^{k-1} \langle v_k, q_j \rangle q_j, \qquad q_k = \frac{u_k}{\|u_k\|}.$$

This is the orthogonal projection onto the orthogonal complement of the span of the previous vectors, followed by normalization.

Orthogonality: Subspaces and Projections

The orthogonal complement of a subspace $W \leq V$ is

$$W^\perp = \{v \in V : \langle v, w \rangle = 0 \text{ for all } w \in W\}.$$

This is a subspace, $\dim W + \dim W^\perp = \dim V$, $(W^\perp)^\perp = W$, and $V = W \oplus W^\perp$ (orthogonal direct sum).

The orthogonal projection $P_W : V \to W$ sends each v to its nearest point in W (in the inner product norm). It is characterized by $P_W v \in W$ and $v - P_W v \in W^\perp$. In terms of an orthonormal basis $\{q_1, \ldots, q_k\}$ for W:

$$P_W v = \sum_{i=1}^{k} \langle v, q_i \rangle q_i.$$

As a matrix: $P = QQ^T$ where $Q = [q_1 \mid \cdots \mid q_k]$. Projections are idempotent ($P^2 = P$) and self-adjoint ($P^T = P$).

The best approximation theorem: $P_W v$ is the closest vector in W to v, i.e., $\|v - P_W v\| \leq \|v - w\|$ for all $w \in W$. This underpins least-squares regression, Fourier analysis, and signal decomposition.

Symmetric Matrices and the Spectral Theorem

A matrix A is symmetric if $A = A^T$ (equivalently, $A_{ij} = A_{ji}$). Symmetric matrices are the real analogues of Hermitian operators and have especially clean eigentheory.

The real spectral theorem: if A is a real symmetric $n \times n$ matrix, then:

1. All eigenvalues of A are real.
2. Eigenvectors corresponding to distinct eigenvalues are orthogonal.
3. A is orthogonally diagonalizable: there exists an orthogonal matrix Q (with $Q^{-1} = Q^T$) such that $Q^T A Q = D$ is diagonal.

Proof of orthogonality of eigenvectors: suppose $Av = \lambda v$ and $Aw = \mu w$ with $\lambda \neq \mu$. Then

$$\lambda \langle v, w \rangle = \langle Av, w \rangle = \langle v, A^T w \rangle = \langle v, Aw \rangle = \mu \langle v, w \rangle.$$

Since $\lambda \neq \mu$, we conclude $\langle v, w \rangle = 0$.

The spectral decomposition for symmetric matrices: $A = \sum_{i=1}^{n} \lambda_i q_i q_i^T$, where $\{q_i\}$ is an orthonormal eigenbasis. Each $q_i q_i^T$ is the orthogonal projection onto the span of q_i.

Eigenvalues and Determinants: The Structural Connection

The determinant $\det(A)$ equals the product of all eigenvalues (over $\mathbb{C}$, counted with algebraic multiplicity):

$$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

This follows because $\det(A) = p_A(0) = \prod_i (0-\lambda_i)\cdot(-1)^n = \prod_i \lambda_i$ (adjusting for the sign convention).

The trace $\mathrm{tr}(A) = \sum_{i=1}^n A_{ii}$ equals the sum of eigenvalues:

$$\mathrm{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$$

Both trace and determinant are invariants under similarity: if $B = P^{-1}AP$, then $\mathrm{tr}(B) = \mathrm{tr}(A)$ and $\det(B) = \det(A)$. The entire characteristic polynomial is a similarity invariant.

A matrix is invertible iff 0 is not an eigenvalue (since $\det(A) = 0$ iff some $\lambda_i = 0$). The inverse has eigenvalues $\{1/\lambda_i\}$ with the same eigenvectors.

For a matrix A with eigenvalue λ and eigenvector v: $A^k v = \lambda^k v$, and if f is any polynomial, $f(A)v = f(\lambda)v$. This is the key computational leverage of eigenvectors: they diagonalize polynomial functions of the matrix.

Generalized Eigenvectors and Jordan Normal Form

When A is not diagonalizable (some $m_g(\lambda) < m_a(\lambda)$), there is still a canonical form: the Jordan normal form. A generalized eigenvector of rank k satisfies

$$(A - \lambda I)^k v = \mathbf{0}, \quad (A - \lambda I)^{k-1} v \neq \mathbf{0}.$$

For each eigenvalue λ, the generalized eigenspace $\ker(A - \lambda I)^{m_a(\lambda)}$ has dimension $m_a(\lambda)$. Together the generalized eigenspaces decompose $\mathbb{C}^n$ into a direct sum (the primary decomposition).

The Jordan normal form J of A is block diagonal, with one Jordan block per "chain" of generalized eigenvectors:

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix} \in \mathbb{R}^{k\times k}.$$

Diagonal matrices are Jordan normal forms where every block has size 1.

Matrix Inverse and Its Relationship to Eigenvalues

If A is invertible, then A^{-1} has eigenvalues $1/\lambda_i$ with the same eigenvectors as A. This follows directly: $Av = \lambda v \Rightarrow v = \lambda A^{-1} v \Rightarrow A^{-1} v = (1/\lambda) v$.

The Cayley-Hamilton theorem provides a formula for A^{-1}: since $p_A(A) = 0$ and $p_A(\lambda) = (-1)^n \lambda^n + \cdots + \det(A)$, dividing by $\det(A)$ and solving for I gives

$$A^{-1} = \frac{-1}{\det(A)} \left[(-1)^n A^{n-1} + \cdots + c_1 I\right],$$

expressing A^{-1} as a polynomial in A. This is theoretically important but seldom used for numerical computation.

For the matrix exponential $e^A = \sum_{k=0}^{\infty} A^k/k!$, diagonalization gives $e^A = Pe^D P^{-1}$ where $e^D = \mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})$. The matrix exponential arises in systems of differential equations: the solution to $\dot{x} = Ax$ with $x(0) = x_0$ is $x(t) = e^{tA}x_0$.

Rank, Linear Independence, and the Fundamental Theorem

The fundamental theorem of linear algebra describes four fundamental subspaces of an $m \times n$ matrix A with rank r:

1. Column space $C(A) \leq \mathbb{R}^m$, dimension r.
2. Null space $N(A) \leq \mathbb{R}^n$, dimension $n - r$.
3. Row space $C(A^T) \leq \mathbb{R}^n$, dimension r.
4. Left null space $N(A^T) \leq \mathbb{R}^m$, dimension $m - r$.

The orthogonality relations: $N(A) = C(A^T)^\perp$ and $N(A^T) = C(A)^\perp$. Thus $\mathbb{R}^n = C(A^T) \oplus N(A)$ and $\mathbb{R}^m = C(A) \oplus N(A^T)$ as orthogonal direct sums.

This means: any vector in the input space $\mathbb{R}^n$ decomposes uniquely into a component in the row space (which gets mapped injectively into the column space) and a component in the null space (which gets annihilated). The transformation A is invertible on the row space, with image equal to the column space.

This structural picture explains why rank is the most fundamental invariant of a linear map: it simultaneously measures the dimension of what is preserved and constrains the dimension of what is lost.